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Palatini Variation and the C1 Map

Writing $\omega = \mathring{\omega} + K(T)$ and using the standard algebraic split of $R(e,\omega)$ into $R(\mathring{\omega})$ plus quadratic torsion (total derivatives dropped under A4), variation w.r.t. ω yields purely *algebraic* equations in the irreps $\{T_\mu, S^\mu, q_{\lambda\mu\nu}\}$. With (98),

$$\delta_\omega S_{\text{Stk}} \;\Rightarrow\; S^\mu = 0, \qquad q_{\lambda\mu\nu} = 0, \qquad m_T^2\,(T_\mu - \partial_\mu \epsilon) + \frac{4\alpha_1}{3}\,T_\mu = 0, \tag{99}$$

where α_1 is the coefficient multiplying the T^2 piece coming from $R(e,\omega)$ (fixed by conventions and already used in Sec. 4.1). For $m_T^2 > 0$ this algebraic system has the unique solution

$$\boxed{T_\mu = \partial_\mu \epsilon, \qquad S^\mu = 0, \qquad q_{\lambda\mu\nu} = 0,} \tag{100}$$

which is precisely the C1 *pure-trace alignment* (Sec. 4) written in covector form and implies $\mathcal{T}_\mu = 0$ on-shell.[7]

Two Equivalent Implementations of the *spurion limit*

We isolate the low-energy regime where observables reduce to those used in the main text:

$$\boxed{\begin{array}{l}\text{(i) Hard penalty: } m_T \to \infty \;\Rightarrow\; \mathcal{T}_\mu = 0, \\ \text{(ii) Lagrange current: } \mathcal{L}_\Lambda = \sqrt{-g}\,\Lambda^\mu \mathcal{T}_\mu.\end{array}} \tag{101}$$

In case (i) integrating out ω (equivalently T) produces a functional delta $\delta[\mathcal{T}]$; in case (ii) the constraint is exact already at the classical level. In both realizations, keeping f_ϵ^2 finite leaves a standard $(\partial\epsilon)^2$ spectator; *the spurion posture* used in the main text corresponds to taking $f_\epsilon^2 \to 0^+$ (or treating ϵ as nondynamical) *within the two-derivative truncation*, so that observables depend only on $\partial\epsilon$ through the invariant $\mathcal{T}_\mu$ (and hence through Σ_ϵ once C1 is enforced).

Low-Energy Effective Action (Power Counting)

At energies $E \ll m_T$ one obtains, after eliminating ω,

$$S_{\text{eff}}^{(\text{LO})} = \int d^4x\sqrt{-g}\Big[\tfrac{M_P^2}{2}R(\mathring{\omega}) + \alpha_1\, I_T + \alpha_2\,\Sigma_\epsilon\Big] + \mathcal{O}\Big(\frac{\partial^4}{m_T^2}\Big), \qquad I_T \equiv -\tfrac{1}{4}\,\Pi_{\text{PT}}[T^A{}_{BC}T_A{}^{BC}], \tag{102}$$

and *after* enforcing C1 (Sec. 4), $I_T = -6\eta^2\,\Sigma_\epsilon$, so that the LO bulk reduces to a single invariant line in the $\{I_T, \Sigma_\epsilon\}$ plane (Sec. 3.3).

Appendix G.2. Complementary EFT Pictures (Same Gradient-Only Physics)

Two complementary constructions emphasize that *only* $\partial\epsilon$ can enter observables at this order.

(i) Torsional–Axion Picture

Augment (98) by a topological coupling

$$S_{\text{t-ax}} = \int d^4x\sqrt{-g}\,\frac{\alpha}{4}\,\epsilon\,\text{NY}, \qquad \text{NY} \equiv d\Big(e^A \wedge T_A\Big), \tag{103}$$

which is exact. Under A5, NY is a *boundary convention*; integrating by parts and using (100) gives bulk terms proportional to $\partial_\mu\epsilon$ contracted with improvement currents, hence no new bulk Euler–Lagrange content in the PT-even sector. The low-energy reduction again depends only on Σ_ϵ (plus boundary assignments absorbed into A5).

[7] Equivalently, one may add the quadratic invariant I_T with a positive coefficient and obtain the same irrep alignment. The projector statements of Sec. 2 ensure all scalar contractions are taken in the PT-even, real sector.

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